

STOCHASTIC MODELING OF SPT N-VALUE AND EVALUATION OF PROBABILITY OF LIQUEFACTION AT GUWAHATI CITY

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Standard penetration test (SPT) reveals the spatial complexity of standard penetration resistance (N -value) with depth. In this paper, a 1D stochastic characterization of spatial complexity of N -values is developed by considering data obtained from sixty two boreholes in Guwahati City. The N -value profile is modeled as the sum of deterministic part and a stochastic component. The deterministic part which characterizes the non-stationary mean of the data is determined by linear regression analysis. The remaining error is modeled as a spatial random field. The characterization of error heterogeneity as a homogeneous Gaussian random field successfully captures the observed autocorrelation function. The proposed stochastic model is used to compute the probability of factor of safety against liquefaction by Monte-Carlo simulation. The results obtained are presented in form of fragility surfaces, expressing the probability of liquefaction as a function of magnitude of the earthquake and epicentral distance. It is observed that the probability of liquefaction at Guwahati city due to strong earthquakes occurring even at large distances is very high.

Keywords: SPT; non-stationarity; exponential random field; Guwahati City; liquefaction probability; fragility surface.

1. Introduction

Seed and Idriss [1971] proposed a procedure to evaluate the liquefaction potential of soils using in-situ test parameter (Standard Penetration Test N value i.e. SPT- N). Subsequently, this procedure has been revised and updated [Seed *et al.*, 1985; Idriss and Boulanger, 2006]. These methods are referred to as deterministic methods, wherein, the liquefaction of soil is predicted to occur if the factor of safety (FS), ratio of resistance of soil against liquefaction to the seismic loading, is less than or equal to 1. However, there are many potential sources of uncertainty in the SPT- N value that strongly influences the liquefaction analysis and hence the FS [Cetin *et al.*, 2004; Juang *et al.*, 2001]. Therefore, a probability-based liquefaction

analysis will help in having a better engineering decision. There have been efforts by several investigators in this direction. Through a detailed review of literature, Rackwitz [2000] has brought out the important aspects of the probabilistic modeling of soils such as, standardisation of different models for consistency and comparability, concept of ergodicity, influence of different soil parameters, geological uncertainty, intrinsic site variability, type of model etc. Manolis [2002] has presented a state-of-the-art review of research in the area of stochastic soil dynamics, which is a relatively new concept in the avenue of probabilistic soil modeling. Cetin *et al.* [2002] using the database of post-earthquake field observations and results of in-situ index tests on soil, have developed a stochastic model for probabilistic assessment of the initiation of soil liquefaction. Hwang *et al.* [2004] presented a reliability based method for estimating the liquefaction probability associated with the FS. Popescu *et al.* [2005] have studied the three dimensional effects in seismic liquefaction analysis of stochastically heterogeneous soils. The study indicates that the excess pore water pressure buildup predicted by stochastic analysis, considering variability of soil, is higher than that obtained from deterministic analysis assuming uniform soil mass with properties taken as the average of the variable soil mass.

Among several uncertainties that influence FS, the uncertainty associated with SPT-N values and peak ground accelerations play a major role in estimating liquefaction hazard. Although, peak ground acceleration can be taken as a lognormal random variable, the modeling of uncertainties associated with SPT-N value is highly complex. Moreover, they have to be treated as a random field. In this paper a probabilistic assessment of liquefaction potential considering uncertainty in SPT-N value has been presented by taking Guwahati city as a case study.

Guwahati City lies in the North Eastern (NE) part of India at 26.18°E , 91.75°N . The North East (NE) region of India is considered as one of the most seismically active regions worldwide. Guwahati is the capital city of the state of Assam and is the largest city (population of about 15 millions as per 2001 Census) in the NE part of the country. Since 1897, this region has experienced two great earthquakes of magnitude above 8.0 and several large earthquakes of magnitudes varying between 6.0 and 8.0, incurring severe damage of property and heavy loss of life. The seismotectonic map of Northeastern region of India (as per the Geological survey of India 2000), along with epicenters of strong earthquakes of magnitude (M_w) > 5 , is shown in Fig. 1. Among these events, the 1897 great Assam earthquake, which had its epicenter at a distance of about 60 km. from Guwahati (Fig. 1) caused widespread liquefaction and large scale damage at the Guwahati city. Structural damage due to foundation failures caused by soil liquefaction was reported in several parts of the city. A bridge made of three girders over a small stream in the city had sank down by 0.45 m due to liquefaction induced deformation in the river bed. The road from Sukleswar ghat to Bhorolumukh, running along the bank of river Brahmaputra, developed several liquefaction induced fissures of width ranging from 0.6 m to 1.2 m. The MSK intensity of this event at Guwahati city has been estimated to be VIII [Oldham, 1899; Ambraseys and Bilham, 2003]. Although, Guwahati is located far away from the epicenter of the 1950 great Assam earthquake (M 8.7) and other

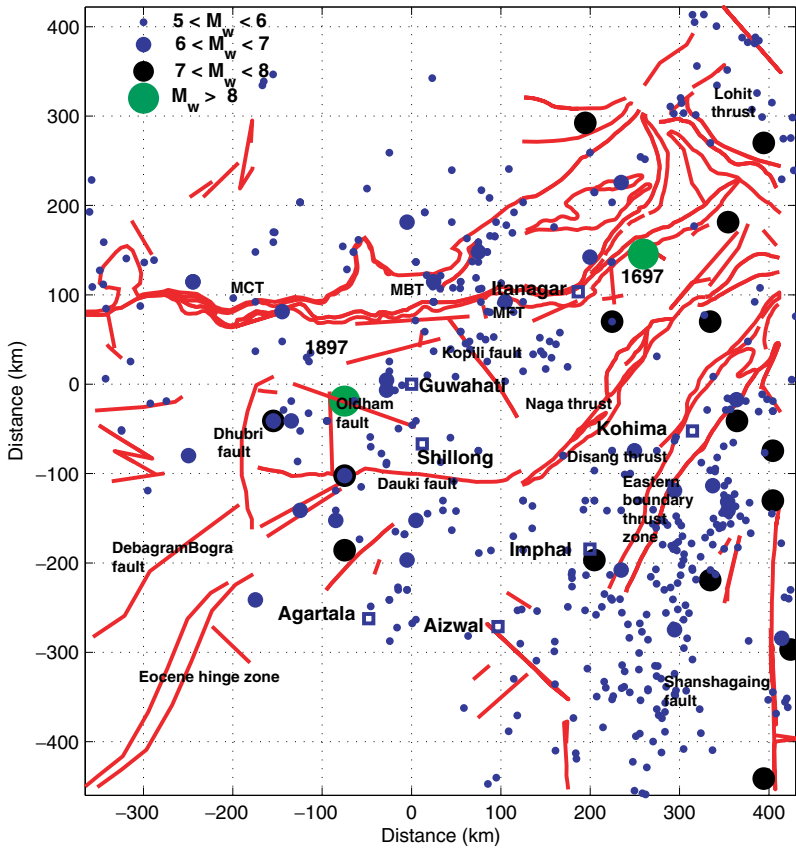


Fig. 1. Seismotectonic map of North Eastern India.

large earthquakes (Fig. 1), even then, non-structural damages have been reported in many parts of the city [Nandy, 2001]. Occurrence of another great earthquake might be highly devastating for Guwahati. Therefore it is of great need to have a reliable estimation of liquefaction hazard due to probable damaging earthquakes occurring in and around Guwahati.

In the Standard Penetration Test, a standard split-barrel sampler is driven into the soil through repeated impact of a standard hammer of 0.63 kN weight, released from a standard height of 0.75 meter and the number of blows (N) required to achieve a standard amount of penetration (i.e. 0.3 m) is measured [IS-2131, 2002]. If the soil is compressible, hence, of low shear strength one would record less value of N while for a relatively incompressible soil that possesses high shear strength the N value will be proportionately higher. Therefore, the SPT- N value is generally correlated with strength parameters of soil. Of recent, it is being widely used in earthquake geotechnical engineering applications, such as, obtaining the shear wave velocity and liquefaction resistance of soil through empirical correlations [Athanasopoulos, 1994; Seed *et al.*, 1985].

At a given site, the SPT is repeated at different depths and locations, for obtaining the spatial variation of N value. In most of the cases the spatial variability of the SPT- N value is highly erratic. This is attributed to the inherent variability of soil caused by geologic process in the past, change in overburden pressure and measurement error due to equipment and operation. The measurement error in case of SPT is significant [Orchant *et al.*, 1988]. Moreover, the obtained N value is not, in fact, a point property but rather local average over some representative volume. Since, SPT is done only at certain discrete depths in the soil; the obtained spatial variation of N values is also not continuous. Apart from randomness, they also exhibit a strong non-stationarity, i.e. an increasing trend with respect to depth.

In this paper, a stochastic model has been developed to simulate the variability of SPT- N values at Guwahati city. In this model the non-stationary part of the data is simulated through linear regression and the error is simulated as an exponential random field. Using this model a large set of SPT- N profiles has been generated for evaluation of the liquefaction probability. Treating the peak ground acceleration as a lognormal random variable, probability of liquefaction is computed by Monte-Carlo simulation. The obtained results, in terms of variation of probability of liquefaction with magnitude and epicentral distance, are presented in form of three dimensional plots, referred to as fragility surface.

2. SPT-N Data

In the present analysis the SPT- N data from 62 boreholes scattered over an area of about 8 km by 8 km over the busy GS road of Guwahati city, its main commercial-administrative hub, are used for developing the model. These data were obtained from several engineering firms involved in the constructions in the Guwahati city. The locations of the boreholes, considered in the present study, along with river Bramhaputra are shown in Fig. 2. The ' N ' values are obtained at intervals of 1.5 m up to a depth of 15 m, except few locations, wherein, it is up to 19.5 m depth. To illustrate the variation of soil and its standard penetration resistance with depth, one sample of soil profile and the corresponding N -value profile at a typical borehole location is shown in Fig. 3. The soil deposit comprises mostly sand, silt and clay (i.e. sand, clay, silt, clayey sand, silty sand, clayey silt, sandy clay, silty clay etc.) in layers. The fines content in the soil is very high (i.e. in range of 65–76%). These are typically alluvial/fluvial deposits.

The N -values show a spatial increasing trend with depth (Fig. 3). The average standard penetration resistance ($\bar{N}$) at a given borehole can be obtained using the following expression as suggested by International Building Code [IBC, 2003].

$$\bar{N} = \frac{\sum_{i=1}^n d_i}{\sum_{i=1}^n \left(\frac{d_i}{N_i} \right)}, \quad (1)$$

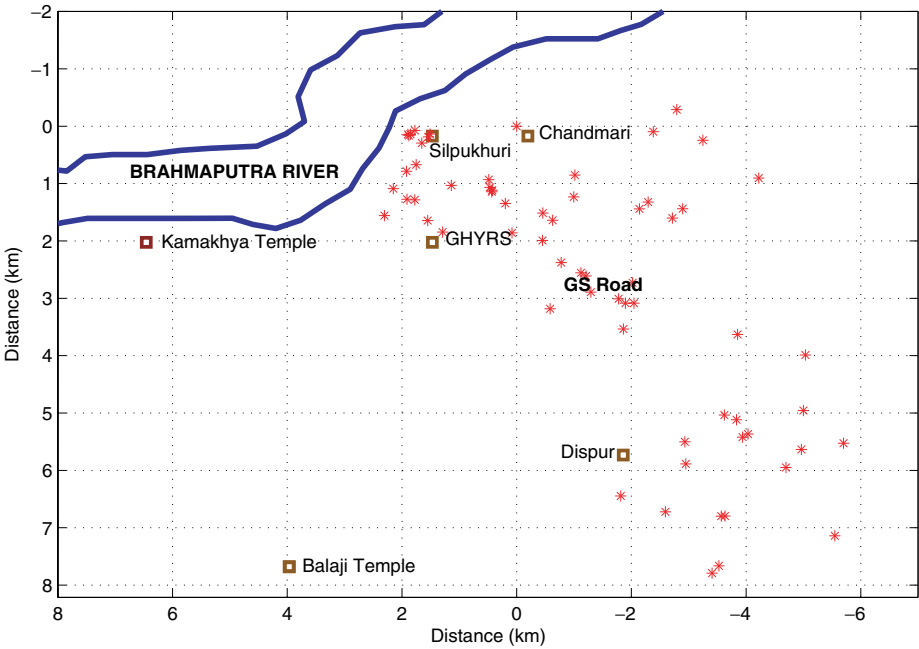


Fig. 2. Locations of boreholes in Guwahati City.

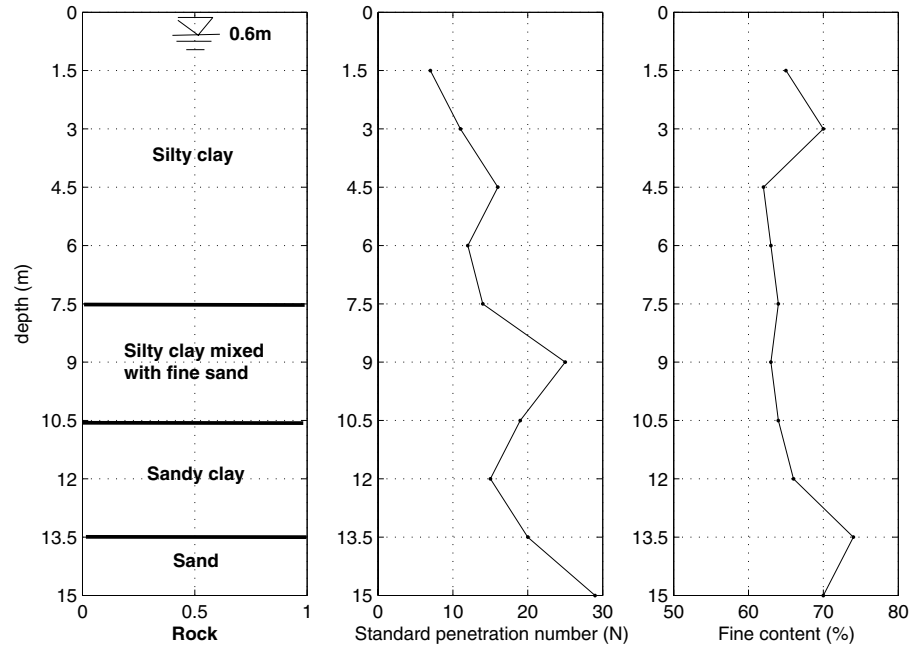


Fig. 3. Typical soil and SPT-N profile in Guwahati City.

where, d_i is the thickness of a given soil layer and n is the total number of layers. When the value of $\bar{N}$ is less than 15, as per [IBC, 2003], the site is classified as E-type site that is highly susceptible to liquefaction. It is observed that the $\bar{N}$ for all the boreholes considered in the present study is less than 15, therefore, all the sites considered in this study are classified as E type and hence are prone to liquefaction.

3. Stochastic Modeling of SPT ‘N’ Value

It is clear from Fig. 3 that, variation of N -value with depth shows an increasing trend, however, it is highly erratic. This non-stationary behavior in the N -value profile is also observed at all other boreholes (Fig. 4(a)). Before proceeding for further statistical analysis, this spatial trend in the data has to be modeled. For this purpose, the measured N values are expressed as the sum of spatial trend (N_{av}) and residual (ε) as

$$N(z) = N_{av}(z) + \varepsilon(z), \quad (2)$$

where, z represents the depth of soil. In the present study, the spatial trend is modeled by linear regression as

$$N_{av}(z) = 3.8208 + 1.1115z. \quad (3)$$

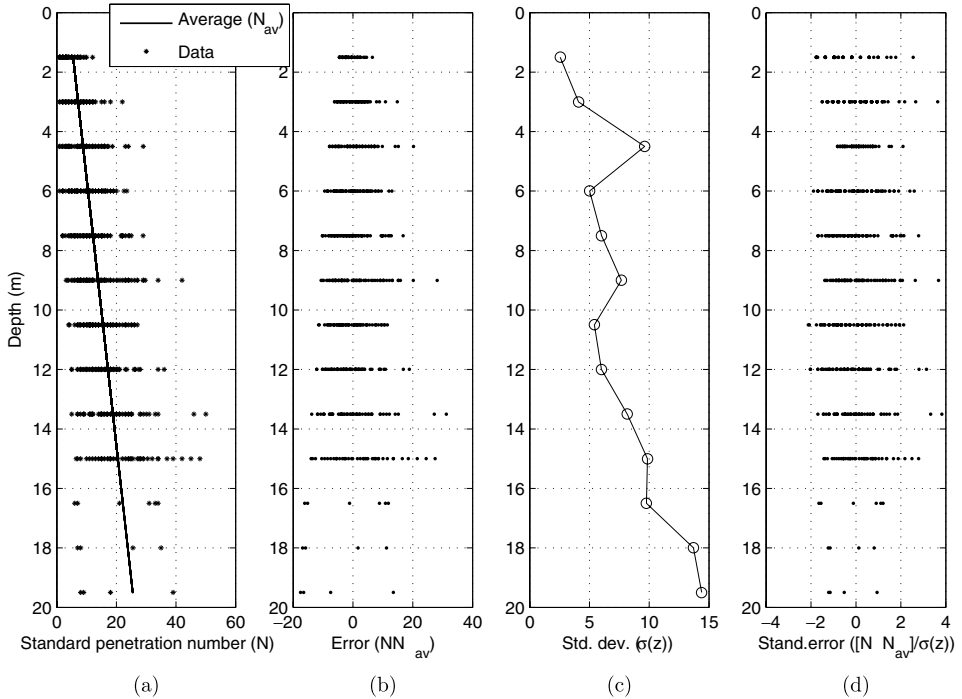


Fig. 4. Standardization of SPT values obtained at Guwahati City.

The coefficients in the above equation have been obtained by evaluating the N values at all the 62 boreholes. The N -values at different depths for all the boreholes along with the fitted trend (Eq. 3) is presented in Fig. 4(a). This equation is used for de-trending the data for further statistical analysis, as discussed below.

3.1. Modeling of residuals

Error ε can be determined by subtracting the N_{av} (Eq. 3) from the original SPT- N data. Once the trend in the data is removed, the remaining part (i.e. ε) can be assumed to be stationary. The spatial distribution of error with depth is shown in Fig. 4(b). This error being highly complex needs a large number of parameters to be modeled through deterministic approaches which is extremely difficult. In such situations, stochastic models provide an alternative for characterizing the complexity of error. This approach requires a very few parameters for simulating the complex variation of data. It is relatively easier to perform the stochastic analysis using standardized data (the zero-mean, unit-variance; $\bar{\varepsilon}$) obtained as

$$\bar{\varepsilon}(z) = \frac{[N(z) - N_{av}(z)]}{\sigma(z)}, \quad (4)$$

where $\sigma(z)$ is the standard deviation of the residuals and is shown in Fig. 4(c). The corresponding normalized residuals are shown in Fig. 4(d). The histogram of normalized residuals is depicted in Fig. 5(a). The skewness and kurtosis of the standard error are also reported in the same figure. The possibility of taking this to be a normal distribution can be verified from the normal probability plot shown in Fig. 5(b). It can be observed that the normalized residuals follow a linear trend

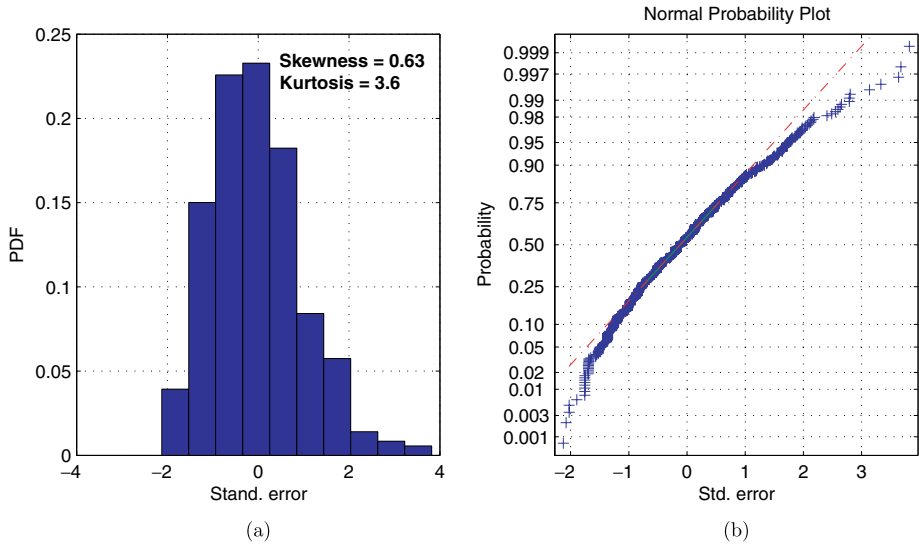


Fig. 5. Probability density function of error.

indicating that the standardized error can be modeled as a normal random variable. These residuals can further be tested for normality using the Kolomogorov-Smirnov (K-S) test [Benjamin and Cornell, 1970]. Application of the K-S test on standardized error yields the statistical value of 0.0504 which is less than the critical value 0.0506 at 95% confidence level. Hence as a first approximation, the normal assumption is accepted for the standardized residuals. In second order analysis, the random spatial variability of a homogeneous random field at two different locations is characterized either in space by an auto-correlation function or in the wave number domain by a power spectral density. Assuming the obtained error field as ergodic, the sample autocorrelation function for a stationary zero-mean, unit-variance random process is computed as

$$\rho(r\Delta z) = \frac{1}{M-r} \sum_{i=1}^{M-r} \bar{\varepsilon}[i\Delta z] \bar{\varepsilon}[(i+r)\Delta z], \quad (5)$$

where $r = 1, 2, \dots, l$; l is the lag distance and $r\Delta z$ is the spatial lag. In Fig. 6 the computed auto-correlation function of the standardized error at all the boreholes is shown. There are several theoretical one-dimensional correlation functions available in the literature. Two such correlation functions widely used in literature are shown in Table 1. The normalized power spectral density $s(\kappa)$, where κ is the wave number, is also shown in the same table.

The parameter, a , of the exponential and squared exponential field has to be estimated from the standardized error by minimizing the mean square error between

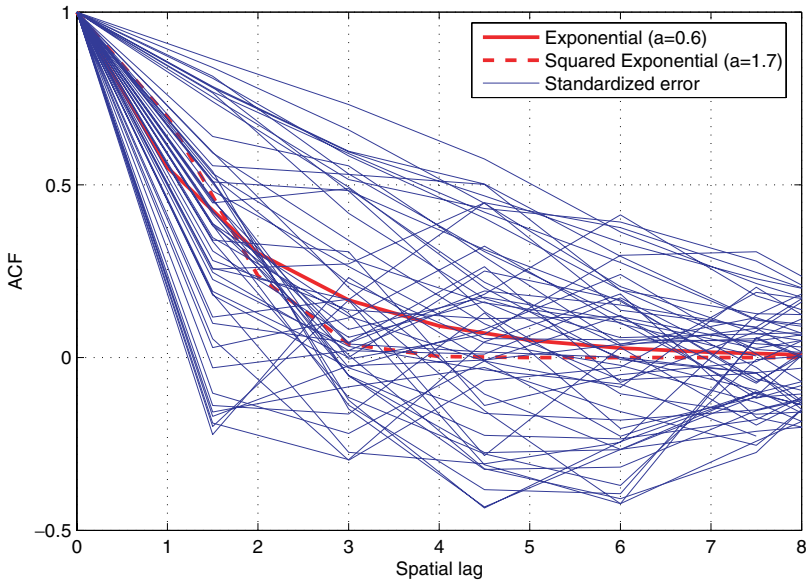


Fig. 6. Auto correlation function (ACF) along with fitted exponential and squared exponential function.

Table 1. One dimensional correlation structures.

Correlation structure	Correlation function $\rho(\xi)$	Correlation distance(θ)	Normalized PSD $s(\kappa)$
Exponential	$e^{-a \xi }, (a > 0)$	$\frac{2}{a}$	$\frac{a}{\pi(a^2 + \kappa^2)}$
Squared exponential	$e^{-(\xi/a)^2}$	$a\sqrt{\pi}$	$\frac{a}{\sqrt{\pi}}e^{-(a\kappa/2)^2}$

the computed auto-correlation and the expressions shown in Table 1. In Fig. 6, the fitted exponential and squared exponential auto-correlation function is plotted. Besides, the estimated parameters are also shown in the same figure. The estimated correlation distances for the exponential and squared exponential are 3.44 and 3.33 respectively. The misfit associated with exponential autocorrelation function is slightly lower than that obtained from squared exponential function and hence the standard error is modeled as an exponential random field for further study.

3.2. Simulation of N -profiles

In the present study, the normalized residuals are simulated as exponential random field. Once the correlation distance is known, the SPT N -value can be simulated from the power spectral density as a Gaussian homogeneous field as

$$N(z) = N_{av}(z) + \sigma(z)\bar{\varepsilon}(z), \quad (6)$$

where $\bar{\varepsilon}(z)$ is a zero-mean, unit-variance homogeneous exponential random field with the power spectral density as given in Table 1. In the present study, the spectral representation method of Shinozuka and Deodatis [1996] is used for simulating moment field. In this method, the power spectral density function is assumed to be insignificant outside the region defined by $-\kappa_u \leq \kappa \leq \kappa_u$, where κ_u is the cut off wave number and is greater than zero. The upper cut-off wave number in the z direction is fixed, depending on the required discretization in the spatial domain, as $\pi/\Delta z$. The discretization in the wave number domain is given by $\Delta\kappa = \kappa_u/m$. Once these parameters are fixed, a one-dimensional homogeneous stochastic field is simulated as

$$\bar{\varepsilon}(z) = \sqrt{2} \sum_{n=0}^{m-1} \sqrt{2s(n\Delta\kappa)\Delta\kappa} \cos(n\Delta\kappa z + \Phi_n), \quad (7)$$

where $\Phi_1, \Phi_2, \dots, \Phi_n$; are random phase angles uniformly distributed between 0 and 2π . In the present study Δz is taken as 1.5 m. From the spectral representation procedure, one can simulate several realizations of the error field by replacing the sequences of random phase angles. To this error field, the non-stationary part computed from Eq. (3) is added for simulating the spatial variation of N -value. The above procedure can be repeated for several realizations of the error field and an ensemble of N -value profiles can be obtained. Several such sample realizations

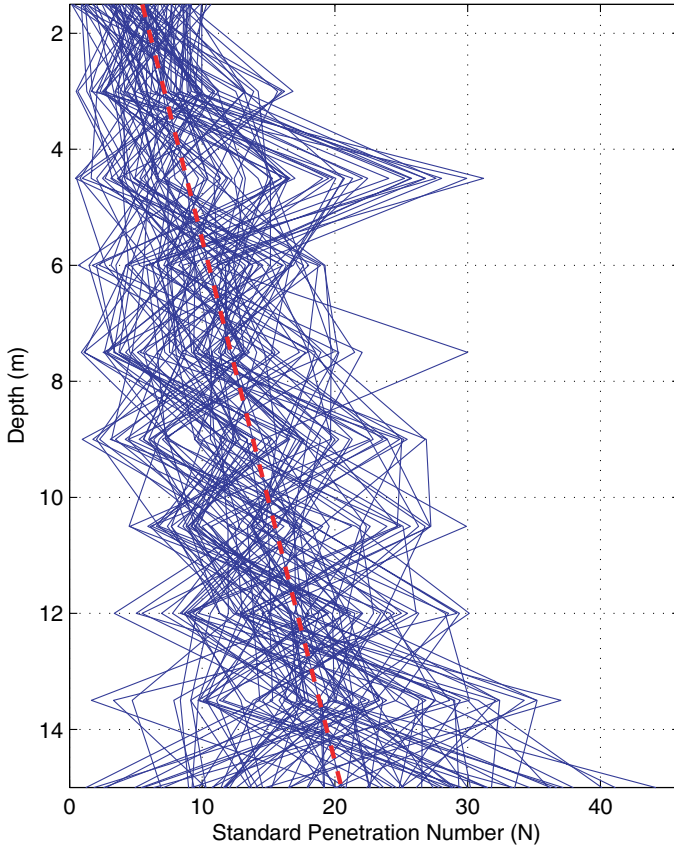


Fig. 7. Sample realizations of SPT-N value.

of the N -values for generalized stochastic model are shown in Fig. 7. It could be observed that the value of N , in the thousand numbers of simulated N -profiles, is in the range of 0–45. This is in agreement with the observed data shown in Fig. 4(a) indicating that the proposed model (trend plus error) is able to replicate the complex trends of the observed data.

4. Generation of Earthquake Ground Motion

Evaluation of liquefaction potential requires peak ground acceleration (a_{max}) during the earthquake. This is generally estimated from the ground motion relation where a_{max} is expressed as function of magnitude and rupture distance. Due to lack of ground motion data during strong earthquakes, no such relation is available for NE India. In the absence of past strong motion data, the only way to estimate a_{max} is to use a ground motion relation developed in a region which belongs to the same tectonic regime as NE India. Recently Raghukanth and Somala [2008]

modeled the available strong motion data of moderate earthquakes with magnitudes less than 6, in NE India. The obtained quality factor which characterizes the regional attenuation indicates that NE region behaves similar to an active tectonic region like California. Through modeling of Coda waves in NE region of India, Gupta and Kumar [2002] also have arrived at similar conclusions. Therefore, in the present study the ground motion relation proposed by Campbell and Bozorgnia [2003] for active tectonic regions is used for estimating peak ground acceleration. This relation has been developed by considering the strong motion data from several earthquakes recorded in seismically and tectonically active, shallow crustal regions located throughout the world. In absence of hanging wall factor, the proposed general attenuation relation of Campbell and Bozorgnia [2003] is given by the expression:

$$\ln a_{max} = c_1 + f_1(M_w) + c_4 \ln \sqrt{f_2(M_w, r, S) + f_3(M_w, r)F + f_3(F) + f_4(S) + \varepsilon}, \quad (8)$$

where M_w is the moment magnitude and r is the closest distance to the fault. ε is a random error term with zero mean and standard deviation equal to $\sigma_{\ln a_{max}}$. The magnitude and distance scaling characteristics are given by the expressions

$$f_1(M_w) = c_2 M_w + c_3 (8.5 - M_w)^2, \quad (9)$$

and

$$f_2(M_w, r, S) = \sqrt{r^2 + g(S)^2 (e^{c_8 M_w + c_9 (8.5 - M_w)^2})^2}, \quad (10)$$

where $g(S)$ characterizes the near-source effect of local site conditions and is given by

$$g(S) = c_5 + c_6 (S_{VFS} + S_{SR}) + c_7 S_{FR}. \quad (11)$$

S_{SR} is the indicator variable for soft rock, S_{FR} is the indicator variable for hard rock, and S_{VFS} is the indicator variable for very firm soil. The style of faulting mechanism is modeled by the expression:

$$f_3(F) = c_{10} F_{RV} + c_{11} F_{TH}, \quad (12)$$

where F_{RV} is the indicator variable for reverse faulting, F_{TH} is the indicator variable for thrust faulting. The far-source effect of local soil conditions is given by

$$f_4(S) = c_{12} S_{VFS} + c_{13} S_{SR} + c_{14} S_{FR}. \quad (13)$$

It is assumed that the soil layer is underlain by generic rock. For computing a_{max} on generic rock, the indicator variables setting S_{SR} and S_{FR} are taken as 0.5 and S_{VFS} as zero. The style of faulting is also not known for several faults in the NE region. For this purpose, the indicator variables F_{RV} and F_{TH} are set as 0.25 in

Eq. 12. The standard deviation of $\ln a_{max}$ is defined in terms of M_w and is given by the expression:

$$\sigma_{\ln a_{max}} = \begin{cases} c_{16} - 0.07M_w & \text{for } M_w < 7.4 \\ c_{16} - 0.518 & \text{for } M_w \geq 7.4 \end{cases} \quad (14)$$

Campbell and Bozorgnia [2003] estimated the coefficients $c_1, c_2, \dots, c_{16}$; by non-linear regression analysis using the method of generalized least squares. A total of 443 accelerograms coming from 36 earthquakes recorded in active regions of the world are used in finding the constants. These constants for horizontal peak ground acceleration are

$$\begin{aligned} c_1 &= -4.033; c_2 = 0.812; c_3 = 0.036; c_4 = -1.061; c_5 = 0.041; c_6 = -0.005; \\ c_7 &= -0.018; c_8 = 0.766; c_9 = 0.034; c_{10} = 0.343; c_{11} = 0.351; c_{12} = -0.123; \\ c_{13} &= -0.138; c_{14} = -0.289; c_{15} = 0.370; c_{16} = 0.920; c_{17} = 0.219. \end{aligned} \quad (15)$$

Once magnitude and distance from the fault are known, mean and standard deviation of the peak horizontal acceleration can be estimated from Eqs. 8 and 15.

5. Evaluation of Liquefaction Potential

Recently Idriss and Boulanger [2006] have proposed an updated and improved method for evaluation of liquefaction potential of soil deposits which has been used in the present analysis. As per this method the earthquake induced loading expressed in terms of cyclic stress ratio, CSR ; is compared with the liquefaction resistance of the soil, expressed in terms of cyclic resistance ratio, CRR . At locations where the loading exceeds the resistance (i.e. Factor of Safety, $FS < 1$), liquefaction is expected to occur. The factor of safety against liquefaction is expressed as

$$FS = \frac{CRR}{(CSR)_{M=7.5, \sigma=1}}. \quad (16)$$

5.1. Cyclic Stress Ratio (CSR)

The cyclic stress ratio (CSR) induced by earthquake ground motions, at a given depth z below the ground surface, is calculated using the following equation [Seed and Idriss, 1971].

$$CSR = 0.65 \frac{a_{max}}{g} \frac{\gamma z}{\sigma'_v} r_d \quad (17)$$

where, a_{max} is the peak horizontal acceleration on the ground surface, γ is the bulk unit weight of soil; σ'_v is the effective overburden pressure, g is the acceleration due to gravity and r_d is the stress reduction factor that measures the attenuation of peak shear stress with depth due to the non-elastic behavior of soil. All the boreholes considered in the present work being less than 34 m depth, the stress reduction

coefficient (r_d) is calculated using the following equations [Golesorkhi, 1989]

$$\ln(r_d) = \alpha(z) + \beta(z)M, \quad (18)$$

$$\alpha(z) = -1.012 - 1.126 \sin\left(\frac{z}{11.73} + 5.133\right), \quad (19)$$

$$\beta(z) = 0.106 + 0.118 \sin\left(\frac{z}{11.28} + 5.142\right). \quad (20)$$

As practiced, the values of CSR that pertain to the equivalent uniform shear stress induced by an earthquake having a moment magnitude M_w , is corrected through the magnitude scaling factor (MSF , Eq. 21), so that the adjusted values of CSR would pertain to the equivalent uniform shear stress induced by an earthquake having a moment magnitude $M_w = 7.5$, i.e. $(CSR)_{M_w=7.5}$.

$$(CSR)_{M_w=7.5} = \frac{CSR}{MSF} = 0.65 \left(\frac{\sigma_v a_{max}}{\sigma'_v g} \right) \frac{r_d}{MSF}. \quad (21)$$

The magnitude scaling factor (MSF) is calculated using the equation proposed by Idriss [1999] as

$$MSF = 6.9 \exp\left(\frac{-M_w}{4}\right) - 0.058 \leq 1.8. \quad (22)$$

Since the liquefaction resistance increases with increasing confining stress, a correction factor, K_σ , is applied (Eq. 24) such that the values of CSR correspond to an equivalent overburden pressure (σ'_v) of 1 atmosphere.

$$(CSR)_{M_w=7.5, \sigma=1} = 0.65 \left(\frac{\sigma_v a_{max}}{\sigma'_v g} \right) \frac{r_d}{MSF} \frac{1}{K_\sigma}, \quad (23)$$

where,

$$K_\sigma = 1 - C_\sigma \ln\left(\frac{\sigma_v}{p_a}\right) \leq 1.0, \quad (24)$$

$$C_\sigma = \frac{1}{18.9 - 2.55\sqrt{(N_1)_{60}}} \leq 0.3. \quad (25)$$

p_a is the atmospheric pressure = 100 kpa. $(N_1)_{60}$ is the corrected SPT-N value, limited to a maximum value of 37.

$$(N_1)_{60} = C_N N_{60}. \quad (26)$$

N_{60} = N value, corrected for the field test procedures and to an average energy ratio of 60% of the SPT hammer.

C_N = factor to normalize N to a common reference effective over burden stress

$$= \left(\frac{p_a}{\sigma'_v} \right)^\alpha \leq 1.7, \quad \text{where } \alpha = 0.784 - 0.0768\sqrt{(N_1)_{60}}. \quad (27)$$

It can be observed from the Eqs. (26) and (27) that $(N_1)_{60}$ and C_N are inter-dependent. Therefore, an iterative procedure is followed to determine $(N_1)_{60}$ and C_N . A series of iterations, were carried out, until the difference between successive iteration values is less than 0.001.

Table 2. Result of Liquefaction analysis for M_w 8.1, PGA = 0.19 g for a typical borehole data

Depth (m)	Fine content (%)	Unit weight (kN/m^3)	Standard penetration number	Factor of safety
1.5	65	19.8	4	0.69
3	71	19.8	6	0.55
4.5	64	19.9	7	0.56
6	64	19.8	8	0.57
7.5	63	19.9	10	0.60
9	64	20	12	0.64
10.5	65	20	14	0.69
12	70	20.3	15	0.69
13.5	75	20	18	0.78
15	76	20	20	0.79

5.2. Cyclic Resistance Ratio (CRR)

Idriss & Boulanger [2006] recommended the following expression to be used for determining the *CRR*, for a cohesionless soil with any fines content

$$CRR = \exp \left\{ \frac{(N_1)_{60cs}}{14.1} + \left(\frac{(N_1)_{60cs}}{126} \right)^2 - \left(\frac{(N_1)_{60cs}}{23.6} \right)^3 + \left(\frac{(N_1)_{60cs}}{25.4} \right)^4 - 2.8 \right\}, \quad (28)$$

$$(N_1)_{60cs} = (N_1)_{60} + \Delta(N_1)_{60}. \quad (29)$$

$\Delta(N_1)_{60}$ is the correction for fines content in percent (*FC*) present in the soil and is given as

$$\Delta(N_1)_{60} = \exp \left(1.63 + \frac{9.7}{FC + 0.1} - \left(\frac{15.7}{FC + 0.1} \right)^2 \right). \quad (30)$$

More details on the procedure and formulations can be found in Idriss and Boulanger [2006]. Table 2 presents a sample calculation for evaluation of factor of safety against liquefaction (*FS*), at a typical borehole considered in the present analysis.

6. Probability of Liquefaction

To cover all possible variations in the SPT-*N* data due to the inherent variability in the soil, a series of synthetic SPT-*N* profiles have been generated using the proposed stochastic model (Eqs. 6 and 7). Making use of these profiles the factor of safety against liquefaction (*FS*) and its probability density function is obtained. It has been observed that for SPT-*N* profiles beyond 1000 numbers further change in the values of probabilities is marginal. In view of this in all subsequent analysis, herein, a thousand number of synthetic SPT-*N* profiles have been considered.

Using these generated SPT-*N* profiles the factor of safety against liquefaction (*FS*) has been evaluated following the procedure of Idriss and Boulanger [2006],

presented in the previous section. It should be mentioned here that, the fines content in soil (FC), bulk unit weight of soil (γ) and position of water table, at different boreholes considered in the present study do not show much variation. This is attributed to their close proximity to each other and being located within a small region. In view of this in the liquefaction analysis for all the simulated SPT- N profiles, these parameters (i.e. bulk unit weight of soil, fines content in soil and position of water table) are treated as deterministic and are taken same as that of the data from a typical borehole presented in Table 2.

The liquefaction analysis, for all the thousand N profiles, has been performed for the 1897 earthquake [$M_w = 8.1, R = 60$ km, Raghukanth *et al.*, 2007]. The mean peak ground acceleration and its standard deviation for this event are estimated using the attenuation relation proposed by Campbell and Bozorgnia [2003]; Eqs. 8 and 15. Using these values (mean peak ground acceleration and its standard deviation) hundred different samples of peak ground acceleration (a_{max}) have been generated through a log-normal distribution. The normalised histogram for the obtained a_{max} (i.e. for the 1897 event) is presented in Fig. 8. The factor of safety against liquefaction (FS), for all the synthetic N profiles, has been computed at every 1.5 m depth till 15 meters. In the liquefaction analysis, for each of these synthetic N profile, the hundred different generated peak ground acceleration (a_{max}) are used for computing the factor of safety (FS). A total of one lakh FS values have been simulated at every 1.5m depth. The obtained factor of safety (FS) profiles for the two typical a_{max} values are presented in Fig. 9. From the FS -profiles the probability distribution function (CDF) of factor of safety (FS) at every 1.5 meter till 15 meters of depth has been computed and are presented in Fig. 10. Using these distribution functions the probability of occurrence of liquefaction [$P(FS < 1)$] at different depths have been computed and are depicted in Fig. 11.

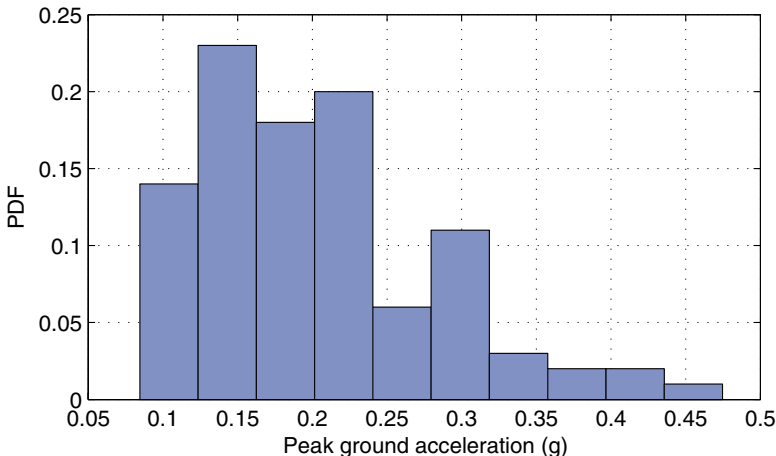


Fig. 8. Normalized histogram of PGA (g) at Guwahati City during the 1897 M_w 8.1 earthquake.

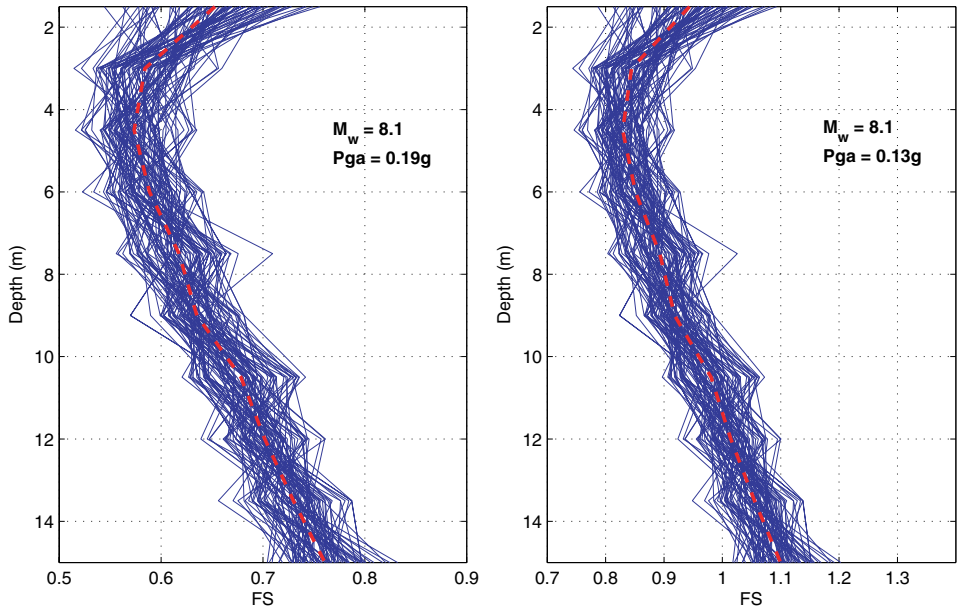


Fig. 9. FS at Guwahati City during the 1897 M_w 8.1 earthquake.

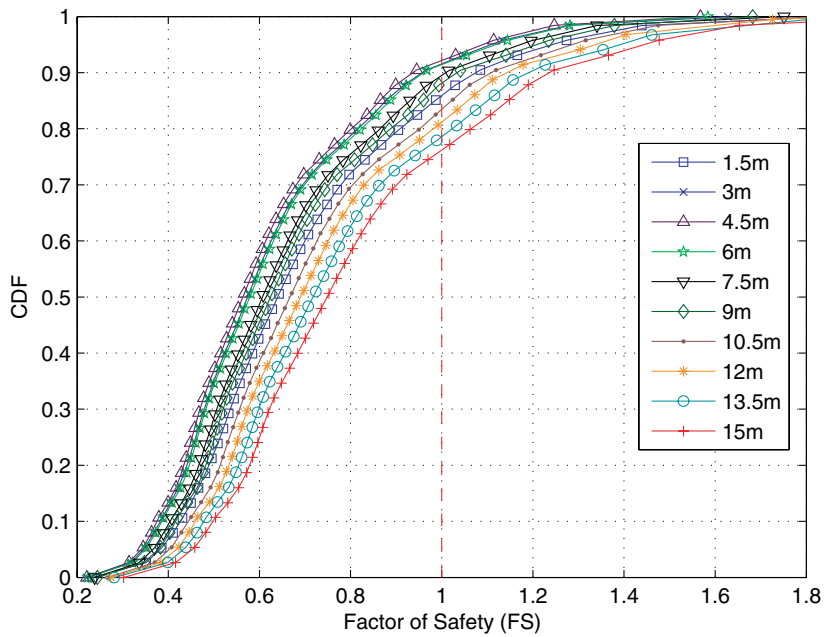


Fig. 10. Probability distribution functions of factor of safety at Guwahati city during the 1897 M_w 8.1 earthquake.

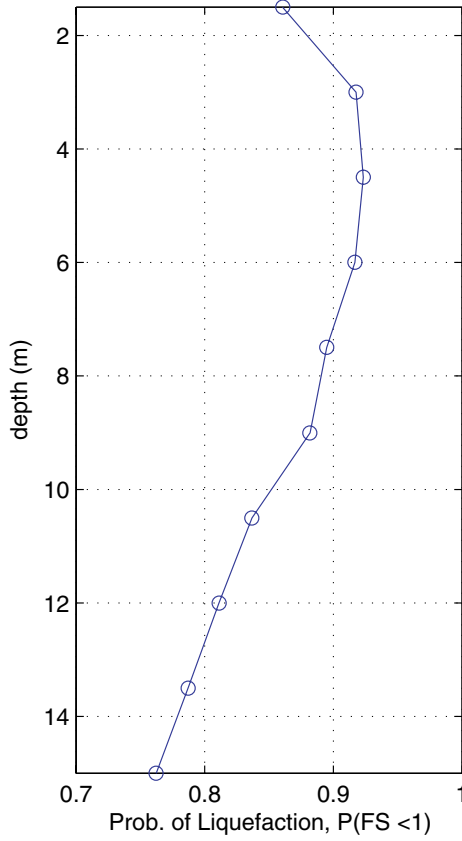


Fig. 11. Probability of liquefaction at Guwahati city during the 1897 M_w 8.1 earthquake.

Figure 11 shows that, for the $M_w = 8.1$ event, the probability of occurrence of the liquefaction (i.e. $FS < 1$) is 0.85 at shallow depth and reduces to 0.77 at a depth of about 15 meters. It is of interest to note that the probability of liquefaction is greater than 0.5 over a depth of about 15 meters from ground surface indicating that the soil is highly susceptible to liquefaction under this event. This is in agreement with the occurrence of widespread liquefaction in the Guwahati City during the 1897 great Assam earthquake [Oldham, 1899].

From the seismotectonic map of North Eastern India, presented in Fig. 1, it could be observed that there are several potential seismic sources in and around Guwahati which can cause severe liquefaction in the city. In view of this, aseismic design of structures in Guwahati city demands evaluation of liquefaction probability over a range of earthquake magnitude and epicentral distances. Therefore, the liquefaction probability analysis for the study area has been performed for earthquake

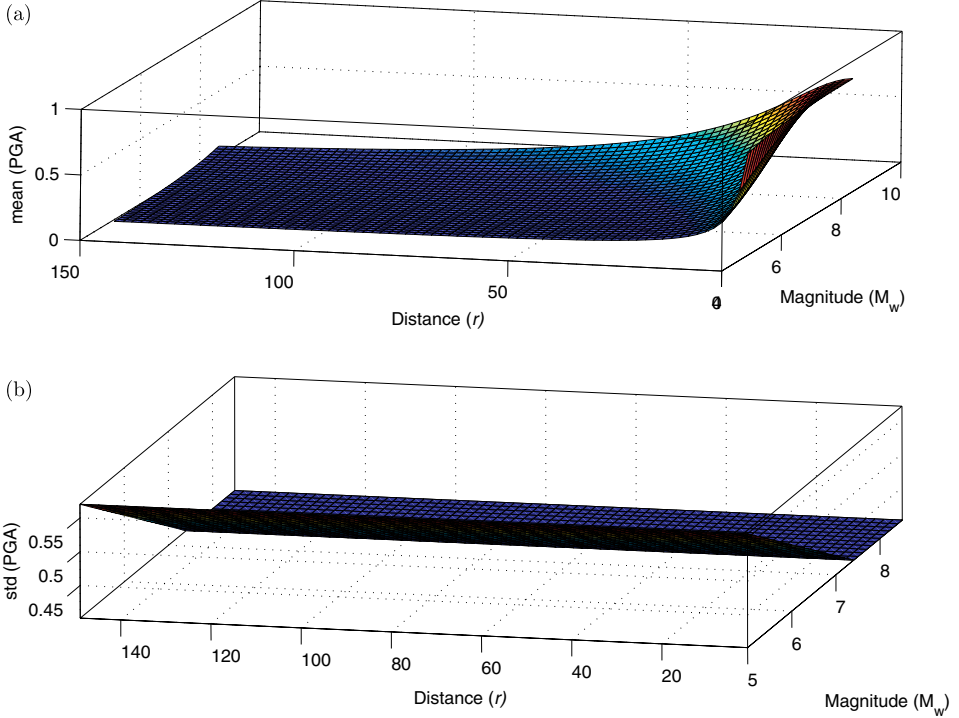


Fig. 12. Mean and standard deviation of peak ground acceleration.

magnitudes (M_w) ranging from 5 to 8.5 at an interval of 0.1 units and the epicentral distance (r) varying from 1 km to 150 km at an interval of 1 km. This leads to a total of 2520 combinations of magnitude (M_w) and epicentral distance (r), wherein, each combination represents a potential seismic source in and around Guwahati city. For each of these cases the mean peak ground acceleration and its standard deviation are estimated using Eqs. 8 and 15. The results are depicted in Figs. 12(a) and 12(b). Using these data the probability of occurrence of liquefaction [$P(FS < 1)$] for all the 2520 cases at different depths has been computed using the procedure mentioned earlier. The results obtained are presented in form of fragility surfaces, in Fig. 13, expressing the probability of liquefaction as a function of magnitude (M_w) and epicentral distance (r). In Fig. 14, the probability of liquefaction, at different depths, at Guwahati City, with distance for two events of magnitude M_w 7 and M_w 8 are shown. From these figures it could be observed that for seismic sources located very near to Guwahati City (i.e. less than about 5 km), irrespective of the magnitude of earthquake the probability of liquefaction is 1 even at large depths as high as 15 meters. For large magnitude earthquakes, even if the seismic source is at a very large distance (as high as 100 km) the probability of liquefaction in the city is still very high [$P(FS < 1) > 0.5$]. It should be mentioned here that seismologists

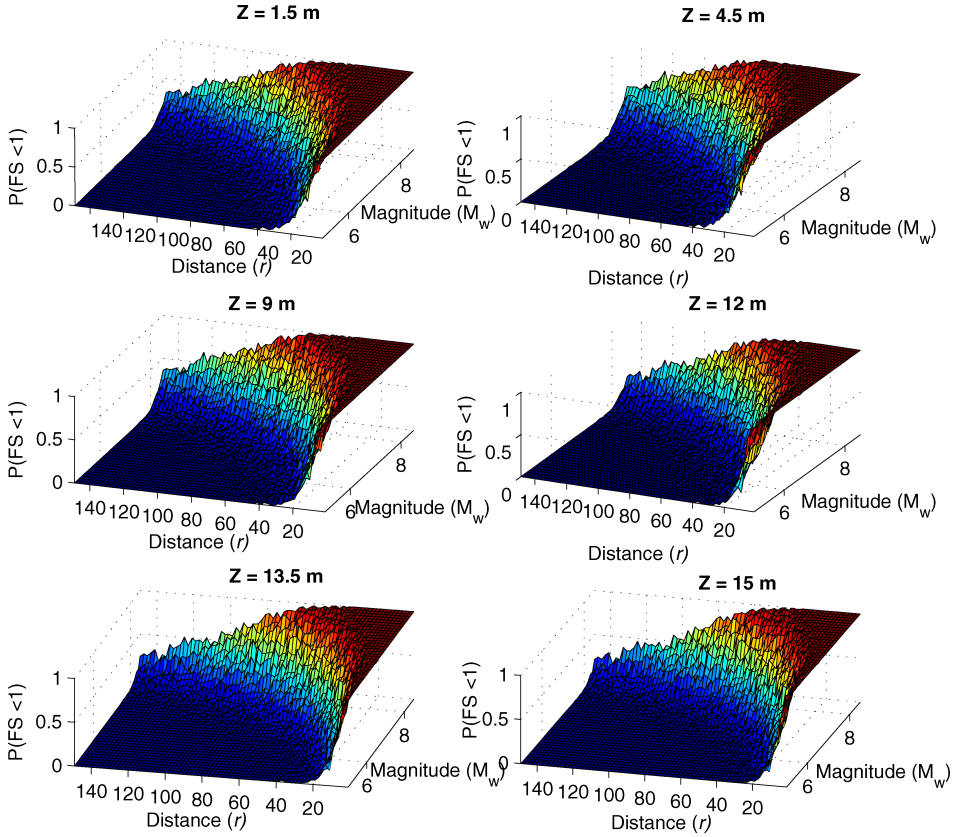


Fig. 13. Probability of liquefaction at Guwahati City as a function of magnitude and rupture distance.

[Khatri and Wyss, 1980; Kayal *et al.*, 2006] are of the opinion that the Himalayan region between the epicenter of 1897 and 1950 earthquake (Fig. 1) is due for a large earthquake of magnitude greater than 8. Guwahati City lies at about 100–200 km from this zone. Hence it could be said that, in case of this event, the city is highly vulnerable to liquefaction.

In a given design, once the location of the fault and the magnitude of the possible earthquake are known, the probability of liquefaction of the site can be directly obtained from the fragility surface (Fig. 13). Hence the presented results in form of fragility surfaces will be of great use for the designers. These surfaces can be used to estimate the probability that the site will be liquefied, in case of a seismic event. However, it should be mentioned here that the deterministic method of Idriss and Boulanger [2006] used in the present analysis is highly conservative [Juang *et al.*, 2002; 2006]. This might have given rise to the relatively higher value of probability of liquefaction (Fig. 13).

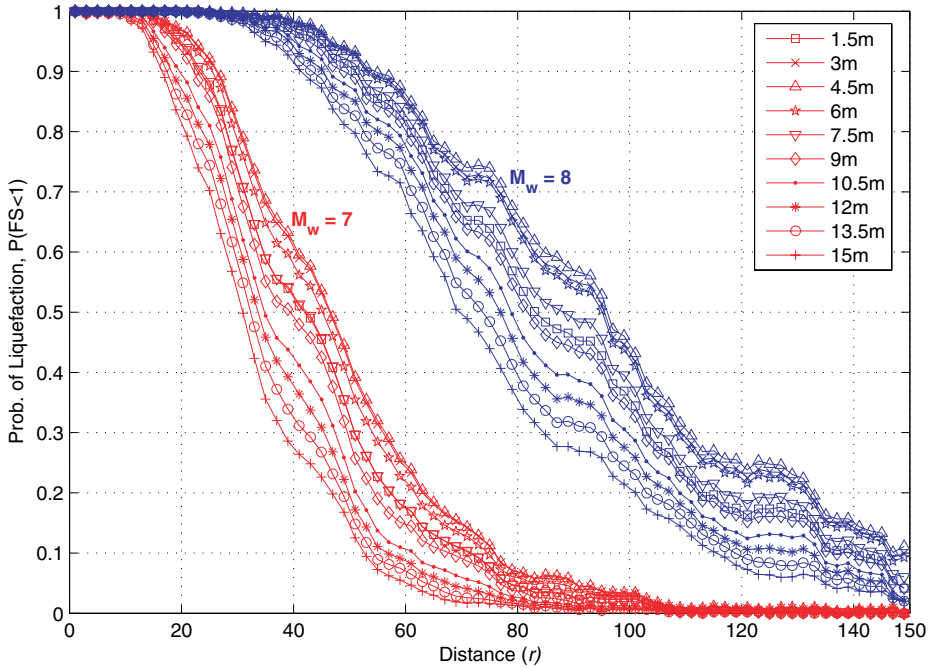


Fig. 14. Probability of liquefaction for different depths at Guwahati City as a function of rupture distance for two different events.

7. Summary and Conclusions

The main objective of the present paper is to develop an approach to characterize the spatial complexity of N -values with depth obtained from SPT. The estimated N -value from SPT at a borehole is taken as the sum of a deterministic part and stochastic component. The proposed stochastic model has been demonstrated by taking borehole data from Guwahati City as a case study. A total of 62 N -profiles obtained from various parts of the city are used for developing the model. The deterministic part, which characterizes the non-stationary mean, is obtained from the data by linear regression analysis. The error in the regression is modeled as a random field. The correlation length of the error field has been estimated from the data. From the obtained correlation length, the error is simulated through the spectral density function as a Gaussian homogeneous random field. Spectral synthesis method is used for generating sample realizations of random field. To this error field the non-stationary part, obtained from Eq. (2), is added for simulating the spatial variation of N -value. Hence the proposed model can be taken as a generalized representation of N -values in Guwahati City. Using the proposed stochastic model the probability of occurrence of liquefaction has been estimated at different depths by Monte Carlo simulation.

Guwahati is the capital city of the state of Assam, and the largest commercial city of North Eastern (NE) region of India which is one of the largest earthquake prone zones of the world. In the past several earthquakes of very high magnitude have caused severe damage to this city. It falls under zone V [IS, 1893] that is the zone of most severe earthquake hazard in the Indian subcontinent. Therefore the findings from the present study will be of immense help in aseismic design and construction of engineering structures. It should be mentioned here that the stochastic model developed in the present study, apart from the liquefaction probability analysis presented herein, can also be used for reliability analysis of geotechnical structures for E type sites in Guwahati City.

In the present study, the stochastic modeling approach has been applied for simulating the N -value profile in Guwahati City. However, the proposed methodology is general and can be used for modeling other soil properties such as shear wave velocity, P-wave velocity etc.

In this paper, uncertainties associated with SPT- N and peak ground acceleration has been considered in estimating liquefaction probability. The fines content in soil, bulk unit weight of soil, are assumed as deterministic variables in this study. Modeling the uncertainties associated with these variables and their effect on liquefaction hazard has to be investigated further.

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